



# Clinical Simulation: An Interactive Simulation Platform for Clinical Reasoning in Emergency Medicine

**Course: Educational Technologies (SoSe 2026)**

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# Clinical Simulation: Project Scope & Team Roles

- **Project Goal:** Interactive web simulation to train medical students in clinical reasoning and cost-effective decision-making.
- **Core Vision:** Shift learning from passive factual memorization to active, inquiry-based clinical problem-solving.
- **Tech Design:** Zero-install, highly responsive, and accessible on any device.

## Team Responsibility

**R. Hefner**

Visual layout, responsive UI/UX, and clinical case writing.

**F. Hauptmann**

Core simulation mechanics, emergency room workflow, and difficulty engine.

**J. Krauß**

Dual-axis assessment, analytics dashboard, and the 10-trial block reflection system.



# Why Do Medical Students Need Simulation Environments?

## The "Clinical Reasoning" Gap

Traditional lectures focus on passive facts. Students struggle to apply knowledge under real ER pressure.

## Cognitive Overload

Novices are overwhelmed by diagnostic options, leading to costly "shotgun-testing" patterns.

## The Safe Sandbox

A risk-free environment where students experience economic consequences without risking patient safety.

## The Solution

A realistic ER simulation that scales difficulty and provides structured feedback for deep reflection.



# PBL in Action: Authentic, Ill-Structured Cases

- **Problem-Based Learning (Barrows, 1986):** Learning is anchored in complex, open-ended clinical problems mirroring real-world practice.
- **Chief Complaint:** The student starts with only basic intake details (e.g., *"A 24-year-old male presents to the emergency department ..."*).
- **Self-Directed Inquiry:** No initial diagnostic guidance or multiple-choice lists; the student must formulate their own initial pathology hypothesis.

The screenshot displays a 'Clinical Simulation' interface. At the top, it shows 'Trial 1 / 10', a 'Dashboard' button, and a 'Budget: \$1000' indicator. Below this is a progress bar with four stages: 1. Anamnesis (active), 2. Diagnostics, 3. Diagnosis, and 4. Feedback. The main content area is titled 'Stage 1: Anamnesis' and contains the text: 'A 24-year-old male presents to the emergency department complaining of abdominal pain that started this morning.' Below this, it says 'Select questions to ask the patient:' followed by six input fields with the following prompts: 'Where exactly is the pain located and has it moved?', 'Can you describe the type of pain?', 'Have you experienced any nausea or vomiting?', 'Have you travelled abroad recently?', 'Do you have any pain when urinating?', and 'Do you have any family history of heart disease?'. At the bottom right, there is a 'Proceed to Diagnostics →' button.

# Active Exploration & Structural Scaffolding

- **Cognitive Apprenticeship (Collins et al., 1989):** Providing a situated, realistic environment (the ER) to develop cognitive skills.
- **Structural Scaffolding:** Strict 4-stage workflow (Anamnesis → Diagnostics → Diagnosis → Feedback) in the navigation bar.
- **Active Exploration:** Enforces history-gathering before diagnostics; prevents novices from rushing to expensive tests.
- **Didactic Design:** The student gathers subjective patient symptoms to refine their differential diagnosis list.

The screenshot displays a 'Clinical Simulation' interface. At the top, there's a header with 'Clinical Simulation' and 'Trial 1 / 10'. To the right are buttons for 'Dashboard' and 'Budget: \$1000'. Below the header is a progress bar with four stages: 1. Anamnesis (active), 2. Diagnostics, 3. Diagnosis, and 4. Feedback. The main content area is titled 'Stage 1: Anamnesis' and describes a 24-year-old male with abdominal pain. It prompts the user to 'Select questions to ask the patient:' and lists six questions with their corresponding patient responses. The questions are: 'Where exactly is the pain located and has it moved?' (response: 'It started around my belly button this morning, but now it has moved down to the lower right side.'), 'Can you describe the type of pain?' (response: 'It's a constant, sharp pain. It gets worse when I move or cough.'), 'Have you experienced any nausea or vomiting?' (response: 'Yes, I completely lost my appetite and threw up once.'), 'Have you travelled abroad recently?' (no response), 'Do you have any pain when urinating?' (response: 'No, passing urine is completely normal.'), and 'Do you have any family history of heart disease?' (no response). A 'Proceed to Diagnostics -->' button is at the bottom right.

**Clinical Simulation** Trial 1 / 10 Dashboard Budget: \$1000

1. Anamnesis 2. Diagnostics 3. Diagnosis 4. Feedback

**Stage 1: Anamnesis**

A 24-year-old male presents to the emergency department complaining of abdominal pain that started this morning.

Select questions to ask the patient:

Where exactly is the pain located and has it moved?  
*It started around my belly button this morning, but now it has moved down to the lower right side.*

Can you describe the type of pain?  
*It's a constant, sharp pain. It gets worse when I move or cough.*

Have you experienced any nausea or vomiting?  
*Yes, I completely lost my appetite and threw up once.*

Have you travelled abroad recently?

Do you have any pain when urinating?  
*No, passing urine is completely normal.*

Do you have any family history of heart disease?

Proceed to Diagnostics -->

# Managing Cognitive Load via Boundary Scaffolding

- **Boundary Scaffolding (Sweller, 1988):** Starting diagnostic budget manages cognitive load and sets explicit limits.
- **Combatting "Shotgun-Testing":** Novices tend to order every test available. The budget forces deliberate, hypothesis-driven clinical choices.
- **Active Experimentation (Kolb, 1984):** A penalty-free navigation loop. Students can move back and forth between Anamnesis ↔ Diagnostics to adjust their hypothesis.

**Clinical Simulation** Trial 1 / 10

Dashboard Budget: \$400

1. Anamnesis 2. Diagnostics 3. Diagnosis 4. Feedback

**Stage 2: Diagnostics**

Select clinical tests to order. Keep relevance and costs in mind.

**Physical Examination (Abdomen)** (Cost: \$100)  
*Tenderness in the right lower quadrant (McBurney's point). Positive rebound tenderness.*

**Complete Blood Count (CBC)** (Cost: \$100)  
*Leukocytes:  $14.5 \times 10^9/L$  (Elevated). Neutrophil dominance.*

**Abdominal Ultrasound** (Cost: \$300)  
*Enlarged, non-compressible appendix with a diameter of 8mm.*

**Chest X-Ray** (Cost: \$100)

**Urinalysis** (Cost: \$100)  
*Normal. To rule out UTI.*

← Back to Anamnesis Proceed to Diagnosis →



# Stimulating Articulation & Active Recall

- **Articulation (Collins et al., 1989):** Forcing students to externalize their implicit reasoning and diagnostic hypotheses.
- **Free-Text Entry vs. Recognition:** By using a text input instead of multiple-choice, we test active clinical recall rather than passive recognition.
- **Hypothesis Synthesis:** The student must synthesize anamnesis data and diagnostic findings to articulate a final, definitive medical diagnosis.
- **Clinical Accuracy Check:** The string is parsed and evaluated against target correct answers behind the scenes.

The screenshot displays a 'Clinical Simulation' interface. At the top, it shows 'Clinical Simulation' with a 'Trial 1 / 10' indicator, a 'Dashboard' button, and a 'Budget: \$400' display. Below this is a progress bar with four stages: '1. Anamnesis', '2. Diagnostics', '3. Diagnosis' (which is the active stage), and '4. Feedback'. The main content area for 'Stage 3: Diagnosis' instructs the user to 'Based on the patient history and diagnostic results, please enter your final diagnosis.' It features a text input field with the placeholder text 'e.g. Broken Arm'. At the bottom, there are two buttons: '← Back to Diagnostics' and 'Submit Diagnosis →'.

# Clinical Evaluation: Assessment, Modeling & Feedback

- 1. Summative Assessment (Scriven, 1967):** The final score (out of 100 points) and qualitative grade (e.g. "Excellent", "Good") evaluating overall performance.
- 2. Cognitive Modeling (Collins et al., 1989):** A side-by-side comparison revealing exactly which diagnostics an expert physician would order and the optimal cost.
- 3. Formative Feedback (Black & Wiliam, 1998):** Actionable clinical explanations clarifying both diagnostic correctness and diagnostic testing efficiency.

The screenshot displays a 'Clinical Simulation' interface. At the top, there's a header with 'Clinical Simulation', 'Trial 1 / 10', a 'Dashboard' button, and a 'Budget: \$400' indicator. Below this is a progress bar with four stages: 1. Anamnesis, 2. Diagnostics, 3. Diagnosis, and 4. Feedback. The current stage is 'Stage 4: Feedback & Assessment'. The main content area shows a 'Summative Assessment' box with a score of '100 / 100' and a 'Final Grade: Excellent (Pass)'. Below this is the 'Expert's Path (Cognitive Modeling)' section, which provides a detailed explanation of how an expert would approach the case, including the optimal targeted diagnostics (Physical Examination (Abdomen), Complete Blood Count (CBC), Abdominal Ultrasound, Urinalysis) and the optimal cost (\$600). The 'Formative Feedback' section follows, showing two feedback items: 'Correct Diagnosis (+60 pts)' and 'Diagnostic Efficiency (+40 pts)'. The interface concludes with a 'Next Patient ->' button.

Clinical Simulation Trial 1 / 10 Dashboard Budget: \$400

1. Anamnesis 2. Diagnostics 3. Diagnosis 4. Feedback

**Stage 4: Feedback & Assessment**

Here is the evaluation of your clinical reasoning processes.

**Summative Assessment**

**100 / 100**

Final Grade: Excellent (Pass)

**Expert's Path (Cognitive Modeling):**

How an expert would approach this:

An experienced doctor prioritizes cost-effectiveness and only orders immediately relevant diagnostics. For this case, the optimal targeted diagnostics were:

**Physical Examination (Abdomen), Complete Blood Count (CBC), Abdominal Ultrasound, Urinalysis.**

Optimal Cost: \$600  
Your Expenditure on Irrelevant Tests: \$0

**Formative Feedback:**

**Correct Diagnosis (+60 pts)**

Excellent! The migrating pain to the RLQ, nausea, leukocytosis, and ultrasound findings point directly to acute appendicitis.

Your Submission: "appendicitis"

**Diagnostic Efficiency (+40 pts)**

Excellent diagnostic choices. You avoided unnecessary tests.

Next Patient ->



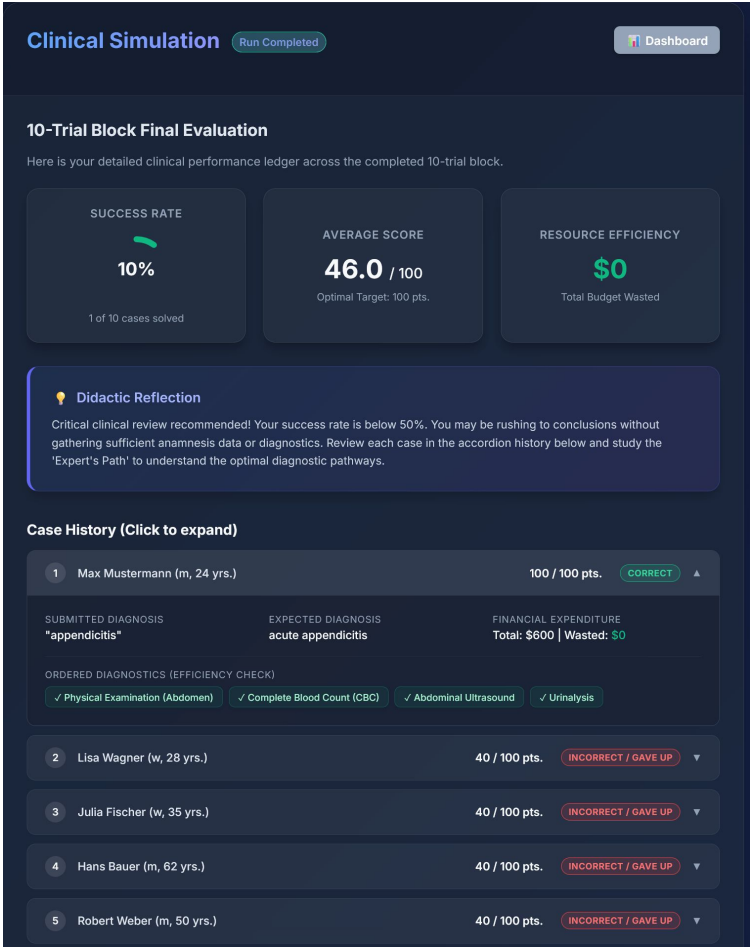
# Metacognitive Reflection & Cognitive Auditing

**Reflection-on-Action (Schön, 1983):** Broadening focus from single-case outcomes to longitudinal cognitive patterns.

**10-Trial Cohorts:** Shuffles and ensures exposure to all 10 clinical conditions in a single block, with state saved across refreshes.

**Didactic Coach:** Analyzes metrics to give targeted metacognitive advice (e.g., warning accurate students about "shotgun-testing").

**Accordion Ledger & Relevance Tags:** Expandable ledger with color-coded tags ([✓ green] for relevant tests, [✗ red] for irrelevant tests) allows students to audit diagnostic search paths.



# Conclusion: A Highly Cohesive Pedagogical Tool

| Educational Concept      | Technical Feature                               | Didactic Outcome                            |
|--------------------------|-------------------------------------------------|---------------------------------------------|
| Problem-Based Learning   | 10 realistic, open-ended cases                  | Cultivates clinical hypothesis formulation  |
| Cognitive Apprenticeship | Situated ER sandbox simulation                  | Makes expert diagnostic workflows explicit  |
| Boundary Scaffolding     | Strict 4-stage UI flow & Budget constraint      | Prevents overload & shotgun-testing         |
| Adaptive Fading          | Budget decreases by \$100 per case solved       | Pushes from exploration to expert precision |
| Articulation & Recall    | Free-text diagnosis input                       | Exercises active recall over passive recog. |
| Formative Feedback       | Actionable clinical explanation & retry options | Enables iterative learning-from-errors      |
| Summative Assessment     | Final score (accuracy & efficiency check)       | Enforces cost-effective clinical care       |
| Learning Analytics       | Persistent modal tracking longitudinal metrics  | Stimulates self-regulated clinical learning |
| Metacognitive Reflection | 10-Trial Block, coaching, tagged ledger         | Promotes systematic reflection-on-action    |

# THANK YOU FOR YOUR ATTENTION!